

Rational interval designs: a general-prime lower bound

Research note; independently reviewed

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Theorem 1. *Suppose rational numbers $x_1, \dots, x_K$ satisfy*

$$\frac{1}{K} \sum_{i=1}^K x_i^j = \frac{1}{j+1} \quad (0 \leq j \leq r).$$

Then, for every prime p ,

$$K \geq p^{\lfloor r/p \rfloor}.$$

In particular $K \geq \max(2^{\lfloor r/2 \rfloor}, 3^{\lfloor r/3 \rfloor})$. If all nodes are p -adically integral, then $p^{\lfloor r/(p-1) \rfloor} \mid K$.

Fix a primitive p th root of unity ζ in an algebraic extension of $\mathbb{Q}_p$, and put $\pi = \zeta - 1$. Write w for the valuation normalized by $w(\pi) = 1$, so $w(q) = (p-1)v_p(q)$ for rational q . The extension $\mathbb{Q}_p(\zeta)/\mathbb{Q}_p$ is totally ramified of degree $p-1$.

Let $G(z) = z/\log(1+z) = \sum_{j \geq 0} g_j z^j$. Then

$$\frac{1}{G(\pi t)} = \sum_{j \geq 0} \frac{(-1)^j \pi^j}{j+1} t^j.$$

All these coefficients are integral for w . Indeed, $j \geq (p-1)v_p(j+1)$ for $j \geq 0$. Equality holds only for $j = 0, p-1$. The cyclotomic equation gives $\pi^{p-1}/p \equiv -1 \pmod{\pi}$, so

$$\frac{1}{G(\pi t)} \equiv 1 - t^{p-1} \pmod{\pi}.$$

Inverting shows

$$w(g_j) \geq -j \quad (j \geq 0), \quad w(g_j) = -j \quad \text{if } p-1 \mid j. \quad (1)$$

Coefficientwise integration identifies $g_j = \int_0^1 \binom{x}{j} dx$. If every node is in $\mathbb{Z}_p$, apply exactness to $\binom{x}{s}$, where $s = (p-1)\lfloor r/(p-1) \rfloor$. Since $\binom{x_i}{s} \in \mathbb{Z}_p$, (1) gives the asserted divisibility of K and hence its lower bound.

Historical attribution. The unit-indicator filter used next is classical. Mit'kin's Lemma 7 [1] gives the cyclotomic-trace polynomial for degrees divisible by $p-1$, with the same congruence precision. His following remark credits the $p=2$ binomial polynomial to Hua (1940) and records Arkhipov's related use in the Hilbert–Kamke problem. The derivation below is included for completeness; the filter itself is not claimed as new. The interval-integral estimate and its application to rational equal-weight quadrature are the specific steps studied in this note; their literature novelty remains unverified.

For the remaining case, define the rational degree- r polynomial

$$H_{r,p}(y) = -\frac{1}{p} \sum_{h=1}^{p-1} \sum_{j=1}^r (\zeta^h - 1)^j \binom{y}{j}.$$

Its coefficients in the binomial basis are integers, since for $j \geq 1$,

$$-\frac{1}{p} \sum_{h=1}^{p-1} (\zeta^h - 1)^j = - \sum_{\substack{0 \leq k \leq j \\ p|k}} (-1)^{j-k} \binom{j}{k}.$$

The root-of-unity filter and the binomial theorem give, first for nonnegative integers y and then by continuity for all $y \in \mathbb{Z}_p$,

$$H_{r,p}(y) \equiv \mathbf{1}_{y \notin p\mathbb{Z}_p} \pmod{p^{\lfloor r/(p-1) \rfloor}}. \quad (2)$$

To see the precision, every omitted term of index $j > r$ has w -valuation at least $j - (p-1)$; the total is rational, so its p -adic valuation is at least $\lceil (r+1)/(p-1) - 1 \rceil = \lfloor r/(p-1) \rfloor$.

Lemma 2. *If $p \mid N$, where N is a positive integer, then*

$$v_p \left(\int_0^1 H_{r,p}(Nx) dx \right) \geq \lfloor r/p \rfloor.$$

Proof. For $\pi_h = \zeta^h - 1$, define

$$W_{N,h}(t) = (1 + \pi_h t)^N - 1.$$

Its degree- k coefficient has w -valuation at least $\lceil (p-1)k/p \rceil + 1$. For $k \geq p$ this follows from the factor π_h^k . For $1 \leq k < p$, the binomial coefficient $\binom{N}{k}$ is divisible by p , which supplies the additional valuation required. Together with (1), this shows that

$$F_N(\pi_h t) := \int_0^1 (1 + \pi_h t)^{Nx} dx = G(W_{N,h}(t)) = \sum_{k \geq 0} A_{k,h} t^k$$

has $A_{0,h} = 1$ and

$$w(A_{k,h}) \geq \left\lceil \frac{p-1}{p}k \right\rceil.$$

This is a formal identity, obtained from $\log((1+z)^N) = N \log(1+z)$. It can also be evaluated at $t = 1$: the terms $g_j W_{N,h}(t)^j$ have valuations at least $(p-1)j$ for integral t , so their sum converges uniformly. Since $p \mid N$, $W_{N,h}(1) = 0$, and therefore $\sum_{k \geq 0} A_{k,h} = 1$. Consequently,

$$\int_0^1 H_{r,p}(Nx) dx = \frac{1}{p} \sum_{h=1}^{p-1} \sum_{k=r+1}^{\infty} A_{k,h}.$$

The w -valuation is at least $\lceil (p-1)(r+1)/p \rceil - (p-1)$. The result is rational, so its p -adic valuation is at least $\lceil (r+1)/p - 1 \rceil = \lfloor r/p \rfloor$. $\square$

Conclusion. Choose the smallest $a \geq 1$ such that every $y_i = p^a x_i$ lies in $\mathbb{Z}_p$. At least one y_i is a unit. Let q count these units, so $1 \leq q \leq K$. By exactness and (2),

$$q \equiv K \int_0^1 H_{r,p}(p^a x) dx \pmod{p^{\lfloor r/(p-1) \rfloor}}.$$

The lemma implies $p^{\lfloor r/p \rfloor} \mid q$, proving the theorem. More precisely, $p^{\min(\lfloor r/(p-1) \rfloor, v_p(K) + \lfloor r/p \rfloor)} \mid q$. $\square$

Scope. No real interval restriction on the rational nodes was used. Repeated nodes are allowed. This proves an exponential lower bound for rational interval designs, but the one-way insertion construction in the dice manuscript does not transfer it to arbitrary dice.

References

- [1] D. A. Mit'kin, *An estimate for the number of terms in the Hilbert–Kamke problem*, Math. USSR Sbornik **57** (1987), no. 2, 561–590, Lemma 7 and its following remark, primary journal record and full text.